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Development Methodology: "Vibe Coding"

A high-velocity, iterative approach to software development that combines AI-assisted synthesis to bridge the gap between abstract concepts and production-ready Luau/TypeScript code.

The iterative efficiency is defined as:

$$\text{Productivity} = \int_{t_0}^{t_{final}} \left(\frac{\text{Prompt Precision} \times \text{Refinement Iterations}}{\text{Architectural Debt}} \right) dt$$

Primary Game & Web Development

CLUTTER (Under Development) | *Lead Developer*

- **Project Scope:** A physics-driven "Growth-Sim" experience built with Luau and Roblox Studio, utilizing Rojo for external workflow synchronization.
- **Key Implementation:** Engineered a volume-retention growth algorithm where the player radius R scales based on the sum of consumed object volumes:

$$R_{new} = \sqrt[3]{R_{old}^3 + \sum V_{objects}}$$

- **Optimization:** Managed a complex asset pipeline from Blender to Studio, utilizing custom rig hierarchies and optimized UV mapping.

Die.io | *Lead Developer (libGDX)*

- **Project Scope:** A Java-based 2D homage to Diep.io developed by a 4-person team.
- **Technical Achievement:** Implemented **Object Pooling** for high-frequency entity spawning (Enemies/Shapes), maintaining a consistent **60 FPS** on mid-range hardware.
- **Architecture:** Designed the core game loop and integrated a backend leaderboard system.

UGCLeaks | *Full-Stack Web Architect*

- **Project Scope:** A central hub for Roblox User-Generated Content (UGC) release schedules.
- **Stack:** Next.js (App Router), PostgreSQL (Neon DB), Serverless Vercel, and Tailwind CSS.
- **Results:** Achieved significant organic traffic within 30 days of launch; implemented "Playground Tools" like the Paintball feature to enhance user retention during high-latency events.

Multidisciplinary Creative Engineering

- **3D Asset Production:** Specialized in low-poly modeling within Blender. Reduced memory overhead through single-texture atlas techniques.
- **Audio-Visual Synthesis:** Transitioned from basic Audacity mashups to complex FL Studio sound design. Leveraged UTAU, Slicer, and Soundfont plug-ins to produce FNF-style "BETADCSI" content with high audio fidelity.
- **Graphic Design:** Designed a comprehensive Memory Album utilizing thematic aesthetics (Nature, Seaside, Daycare), focusing on high-impact visual storytelling and layout composition in Figma.

Technical Specifications

- **Languages & Frameworks:** Luau, TypeScript, Next.js, Java, Node.js, C, PostgreSQL.
- **Tools:** Blender, Roblox Studio, Rojo, Git, FL Studio, Figma.

Looking for graduate-level Game Development studies in Canada.